EE2005 – ELECTRICAL DEVICES AND MACHINES

TUTORIAL 1 – MAGNETIZATION CURVES AND MAGNETIC CIRCUITS

1. A magnetic circuit in the form of a toroid has a mean length of 20 cm and a circular cross-section whose diameter is 2 cm. The core is wound by the coil of 100 turns and carries a current of 5 A. The value of the flux density in the iron is 1.0 T. Calculate the flux in the core, mmf, magnetic field intensity, relative permeability of the magnetic material, and reluctance of the magnetic circuit.

   [Ans: πx10^{-4} Wb; 500 At; 2500 At/m; 318.3; 1,591,549.43 H^{-1}]

2. The iron core in Figure 1 has a mean length $l_c = 100$ cm, uniform cross-sectional area $A_c = 4$ cm^2 and relative permeability $\mu_r = 1500$. The two coils on the core, coil 1 and coil 2, have number of turns $N_1 = 100$ and $N_2 = 50$, respectively. The currents flowing in the respective coils are $I_1 = 2\ A$ and $I_2 = 1.5\ A$ and their directions are indicated. (i) Find the resultant flux produced in the core. (ii) If the direction of I_2 is reversed but the magnitude is unchanged, find the resultant flux produced in the core.

   [Ans: (i) 2.073x$10^{-4}\ Wb$, (ii) 9.425x$10^{-5}\ Wb$]

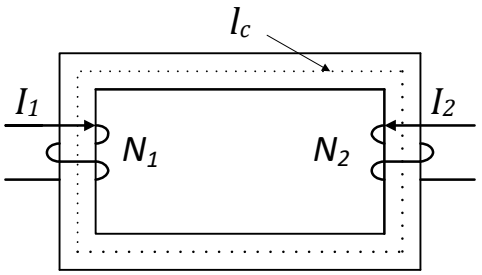


Figure 1

3. Figure 2 shows an electromagnetic system which is made up of three sections.

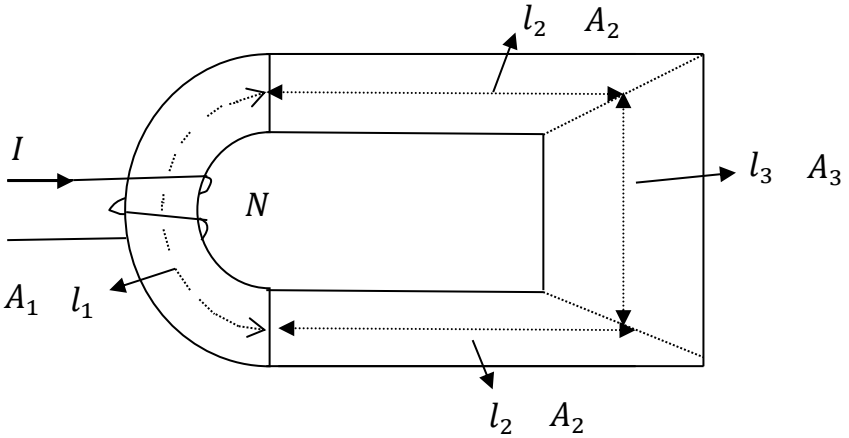


Figure 2

The details are as follows:

$$A_1 = A_2 = 30\text{x}10^{-4}\ m^2, \quad A_3 = 60\ \text{x}10^{-4}\ m^2$$

(cross-sectional areas of the various sections)

$l_1 = 0.225\pi\ m$ (mean length of arc section)

$l_2 = l_3 = 0.45\ m$ (mean lengths of the core sections)

$\mu_1 = 2000,\ \ \mu_2 = \mu_3 = 2500$

(relative permeabilities of the various sections)

$N = 200$ turns, $I = 0.8\ A$

(a) Draw the magnetic equivalent circuit showing the reluctances of various parts.
(b) Determine the flux in the core.
(c) Determine the flux densities in various parts of the core.
(d) Determine the current required in the coil to produce a flux density of 0.4 T in the circular core.

[**Ans: (a) $\mathfrak{R}_{total} = 213117\ H^{-1}$; (b) 0.75 mWb; (c) 0.125 T, 0.25 T; (d) 1.28 A**]

4. The magnetic circuit in Figure 3 has a 200-turn coil wound on the left limb. The cross-sectional areas of the left limb, center limb and the right limb are 12 cm^2, 8 cm^2 and 4cm^2, respectively. Given the following data for the iron circuit, calculate the current in the coil to give a flux of 0.96 mWb in the center limb. Neglect leakage and fringing.

[**Ans: 1.04 A**]

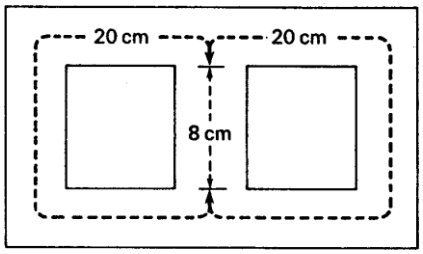


Figure 3

H(At/m)	260	390	500	650	975
B(T)	0.8	0.95	1.02	1.12	1.2

5. Suppose the data for the iron circuit in Q4 is not given. Instead, the relative permeability of the iron $\mu_r = 2450$ is now provided. Calculate the current in the coil to give a flux of 0.64 mWb in the center limb.

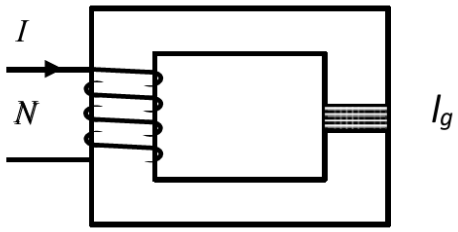
[**Ans: 0.3118 A**]

EE2005 – ELECTRICAL DEVICES AND MACHINES

TUTORIAL 2 – MAGNETIC CIRCUITS AND INDUCTANCE

1. The magnetic structure with an air gap length of $l_g = 0.1$ cm is shown in Figure 1, where the core is wound uniformly with 250 turns. The mean length of the core is 39.9 cm and the cross-sectional area is 8 cm^2. Given the following data for the iron circuit, determine the current *I* needed to produce a flux of 0.72 mWb in the air gap.

[Ans: 3.423 A]



H(At/m)	350	650	975	1200
B(T)	0.9	1.1	1.2	1.25

Figure 1

Suppose the data for the iron circuit is not given, but $\boldsymbol{\mu_r = 2450}$ is provided instead. Calculate the current need to establish at flux of 0.72 mWb in the air gap. Also, find the inductance of the coil.

[**Ans: 3.33 A, 54 mH**]

2. (a) Calculate the inductance of a magnetic toroid core with a 25 cm mean diameter and a circular cross-sectional area 6.25 cm^2. The toroid is wound uniformly with 1000 turns of wire. Find the emf induced when a current increasing at a rate of 200 A/s flows in the winding. The magnetic material has a relative permeability of 800.

[**Ans: 0.8 H; 160 V]**

(b) Suppose a coil of 1000 turns has a flux of 2 μWb linking with it when carrying a current of 2 A. If this current is reduced to 0.5 A in 1 ms, determine the magnitude of the average voltage induced in the coil and the final flux.

[**Ans: 1.5 V; 0.5 μWb**]

3. Consider the magnetic structure of Figure 2 which consists of three sections, namely, the left limb, the center limb and the right limb.

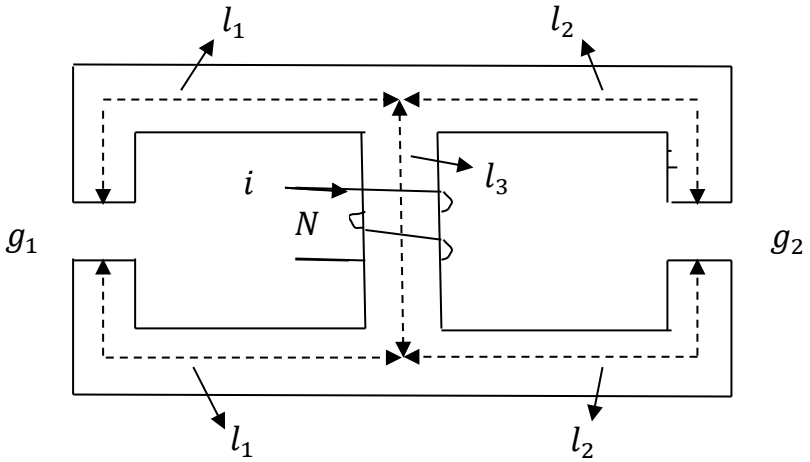


Figure 2

The dimensions of the magnetic structure are as follows:

$l_1 = l_2 = 3.95\ cm$	$l_3 = 4\ cm$	(mean lengths of core)
$A_1 = A_2 = 1\text{x}10^{-4}\ m^2$	$A_3 = 2\text{x}10^{-4}\ m^2$	(cross-sectional areas)
$g_1 = 0.1\ cm$	$g_2 = 0.1\ cm$	(air gap lengths)

The permeability of the magnetic material is $\mu = 2000\mu_o$, where $\mu_o = 4\pi\text{x}10^{-7}$ H/m. The coil consists of $N = 400$ turns and has a total resistance of 1 Ω.

(a) What is the inductance of the system?

[**Ans: (a) 37.92 mH]**

(b) If a dc current of 2.5 A is flowing through the coil, calculate:
   (i) Flux density in the core.
   (ii) The energy stored in the air gaps, the energy stored in the magnetic core and the total energy stored in the magnetic system.

[Ans: (i) 1.185 T; (ii) 0.1117 J, 0.00665 J, 0.1184 J]

(c) If the coil is now connected to a 50 Hz AC source, determine the amplitude of the input voltage required to establish the same flux density in the core as in part (b)(i) above.

[Ans: 29.9 V]